Tutorium - 10:

Satz: (über die Lokale Umkehrbarkeit)
$U \subseteq \mathbb{R}^n$ offen, $f \in C^k(U, \mathbb{R}^n)$ genau dann ein lokaler C^k-Diffeo. falls J_f invertierbar ist.

$$f'(y) = \frac{1}{f'(f^{-1}(y))} \quad \Big| \quad J_{f^{-1}}(f(x)) = (J_f(x))^{-1}$$

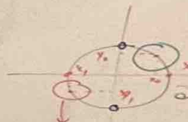
∘ $f: \mathbb{R}^m \times \mathbb{R}^n \longrightarrow \mathbb{R}^n$, $f(x,y) = 0$

$\hat{y}: U \subseteq \mathbb{R}^m \longrightarrow V \subseteq \mathbb{R}^n$ $\qquad y = \hat{y}(x)$

$f: \mathbb{R}^2 \longrightarrow \mathbb{R}$, $f(x,y) = x^2+y^2-1$

$f(x,y) = 0 \Rightarrow x^2+y^2 = 1$

$x_0^2 + y_0^2 = 1 \Rightarrow y_0 = \sqrt{1-x_0^2}$



x_2, y_2

$\partial_y f = 2y = 0$ $\qquad \boxed{\hat{y}(x)}$

$x_2 = \sqrt{1-y_2^2}$

$y_1 = -\sqrt{1-x_1^2}$

V_α

Satz: $D \subseteq \mathbb{R}^{m\times n}$, ① $f \in C^1(D, \mathbb{R}^n)$ $(x_0, y_0) \in D$

② $f(x_0,y_0) = 0$ und $\det(\partial_y f(x_0,y_0)) \neq 0$ ③

$\Rightarrow U \subseteq \mathbb{R}^m$ von x_0 ist y nach x auflösbar.

P10.1: $f: \mathbb{R}^n \times \mathbb{R}^m \longrightarrow \mathbb{R}^m$ linear. $b \in \mathbb{R}^m$ $f(x,y) = b$, $x \in \mathbb{R}^n$,

(i) Wann $f(x,y)$ nach y lösbar. $C = [A \quad B]$

$$f(x,y) = C \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \\ y_1 \\ \vdots \\ y_m \end{pmatrix} = [\underset{\in \mathbb{R}^{m\times n}}{A} \quad \underset{\in \mathbb{R}^{n \times m}}{B}] \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^m = Ax + By = b$$

$$By = b - Ax \quad ; \quad \det B \neq 0$$

(ii) $g: \mathbb{R}^n \longrightarrow \mathbb{R}^m$ | $y = B^{-1}(b - Ax)$;

$x \mapsto B^{-1}(b - Ax)$

$$f(x, g(x)) = Ax + Bg(x) = Ax + \underbrace{B B^{-1}}_{I}(b - Ax) = Ax + b - Ax = b$$

P10.3: $V_\alpha(x) = (x^2-1)^2 - \alpha x$

(a) für $\alpha = 0$: $V_0(x) = (x^2-1)^2$

$V_0'(x) = 4x(x^2-1) \longrightarrow =0$ notwendig!

$V_0''(x) = 4(3x^2 - 1)$ > 0 : Min, < 0 : Max

$V_0'(x) = 4x(x^2-1) = 0 \Rightarrow x_1 = -1, \; x_2 = 0, \; x_3 = +1$

$V_0''(x_{1,3}) > 0 \Rightarrow x_1, x_3$ sind lokale Minima

$V_0''(x_2) < 0 \Rightarrow x_2$ ist ein lok. Maximum

(b) $\alpha \neq 0$, $V_\alpha(x) = (x^2-1)^2 - \alpha x$

$V_\alpha'(x) = 4x(x^2-1) - \alpha = f(\alpha, x)$ $\quad 0 \mapsto \alpha$

$f(0,x)$, x_1, x_2, x_3 sind die Lös.

für bel. (aber klein) α, $\check{x}_i(\alpha)$ lokal nach x_i auflösen.

$V_\alpha''(\check{x}_i(\alpha)) = 4(3\,\check{x}_i(\alpha)^2 - 1)$

$V_\alpha''(\check{x}_1(\alpha)) = \text{—}$

$V_\alpha''(\check{x}_2(\alpha)) = \text{—}$

$V_\alpha''(\check{x}_3(\alpha)) \text{—}$

$$f(x) := \begin{cases} \left(x \sin(\tfrac{1}{x}), |x|^{x^2}\right), & x \in \mathbb{R}\setminus\{0\} \\ (0,1), & x = 0 \end{cases}$$

$f: \mathbb{R} \to \mathbb{R}^2$, Stetigkeit und (partielle) Diffbarkeit

$\nabla f, J_f(x)$

$\lim_{x\to 0} x \cdot \sin(\frac{1}{x}) = 0$

$|\sin\frac{1}{x}| \le 1$

$|x| \quad -|x|$

$\lim_{h\to 0} \frac{f(h)-f(0)}{h}$

$\lim_{x\to 0} |x|^{x^2} = \lim_{x\to 0} e^{x^2 \ln|x|} \stackrel{St.}{=} e^{\lim_{x\to 0} x^2 \cdot \ln|x|} \to e^0 = 1.$

$f(x) = \begin{pmatrix} f_1(x) \\ f_2(x) \end{pmatrix} = \begin{pmatrix} x\sin(1/x) \\ |x|^{x^2} \end{pmatrix}, \mathbb{R}\setminus\{0\}$ $\lim_{x\to 0} f(x) = (0,1) = f(0)$

$\lim_{h\to 0} \frac{f_1(h) - f_1(0)}{h} = \lim_{h\to 0} \frac{h \sin(1/h) - 0}{h} = \lim_{h\to 0} \sin(1/h)$ P.D. ↯

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notwendig!

$V_0'(x) = 4x(x^2-1) \to = 0$

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(b) $\alpha \neq 0$, $V_\alpha(x) = (x^2-1)^2 - \alpha x$

$V_\alpha'(x) = 4x(x^2-1) - \alpha = f(\alpha, x)$ $\quad 0 \mapsto \alpha$

$f(0,x)$, x_1, x_2, x_3 sind die Lös.

für bel. (aber klein) α, $\tilde{x}_i(\alpha)$ lokal nach x_i auflösen.

$V_\alpha''(\tilde{x}_i(\alpha)) = 4(3\tilde{x}_i(\alpha)^2 - 1)$

$V_\alpha''(\tilde{x}_1(\alpha)) = -$

$V_\alpha''(\tilde{x}_2(\alpha)) = -$

$V_\alpha''(\tilde{x}_3(\alpha)) - -$

$x\|x\|$ $\quad Df(0) = 0$

$\frac{x^2}{\|x\|} + \|x\|$ $\quad \mathbb{R}^n \setminus \{0\} \cdot \mathbb{R}$

Tutorium - 10:

x_1, x_2, x_3 $\quad f(\alpha, x) = 4x(x^2-1) - \alpha$

$f \in C^1$ ✓ $\quad f(\alpha, x) = V_\alpha'(\check{x}_i(\alpha)) \overset{!}{=} 0$

$$\check{x}_i'(\alpha) = -\left[\partial_x f(\alpha, \check{x}_i(\alpha))\right]^{-1} \partial_\alpha f(\alpha, \check{x}_i(\alpha))$$

$$= \frac{+1}{12\check{x}_i(\alpha)^2 - 4}$$

$$\check{x}_i''(\alpha) = \frac{d}{d\alpha}\check{x}_i'(\alpha) = \frac{-24\,\check{x}_i(\alpha)\,\check{x}_i'(\alpha)}{(12\check{x}_i(\alpha)^2 - 4)^2} = \frac{-24\,\check{x}_i(\alpha)}{(12\check{x}_i(\alpha)^2 - 4)^3}$$

$$\check{x}_1(\alpha) = -1 + \frac{1}{8}\alpha - \overbrace{\frac{24 \cdot (-1)}{(12(-1)^2 - 4)^3}}^{= 3/128} + O(\alpha^2)$$

$$\check{x}_2(\alpha) = 0 - \frac{1}{4}\alpha + 0 \cdot \alpha^2 + O(\alpha^2)$$

$$\check{x}_3(\alpha) = 1 + \frac{1}{8}\alpha - \frac{3}{128}\alpha^2 + O(\alpha^2)$$

c) $V_\alpha(\tilde{x}_i(\alpha)) = g(\alpha, \check{x}_i(\alpha))$, $i \in \{1,2,3\}$ $\quad x_i \longrightarrow \tilde{x}_i(\alpha)$

$$\frac{d}{d\alpha} g(\alpha, \check{x}_i(\alpha)) = \underbrace{\frac{\partial}{\partial\alpha} g(\alpha, \check{x}_i(\alpha))}_{= -\check{x}_i(\alpha)} + \underbrace{\cancel{\frac{\partial}{\partial x} g(\alpha, \check{x}_i(\alpha))}}_{V'(\check{x}_i(0)) = 0} \frac{\partial}{\partial\alpha}\check{x}_i(\alpha)$$

$$\frac{\partial^2}{\partial\alpha^2} g(\alpha, \check{x}_i(\alpha)) = \frac{d}{d\alpha}(-\check{x}_i(\alpha)) = -\check{x}_i'(\alpha) = \frac{-1}{12\check{x}_i(\alpha)^2 - 4} = \frac{1}{V_0''(\check{x}_i(\alpha))}$$

$$V_\alpha(\check{x}_i(\alpha)) = V_0(x_i) - \alpha \cdot x_i - \frac{1}{2}\alpha^2 x_i'(0)$$

$$V_\alpha(\tilde{x}_1(\alpha)) = \alpha - \frac{1}{16}\alpha^2 + O(\alpha^2)$$

$$V_\alpha(\tilde{x}_2(\alpha)) = 1 + \frac{1}{8}\alpha^2 + O(\alpha^2)$$

$$V_\alpha(\check{x}_3(\alpha)) = -\alpha - \frac{1}{16}\alpha^2$$